



Enhancing road safety in internet of vehicles using deep learning approach for real-time accident prediction and prevention

Xu Wei

Sichuan Top Information Technology Vocational College, Chengdu, Sichuan, 611731, China

ARTICLE INFO

Keywords:

Internet of vehicles
Internet of Things
Deep learning
LSTM
Road-side units
Accuracy

ABSTRACT

The paper proposes an Internet of Vehicles (IoV)-based Accident Prediction and Prevention System that leverages the Internet of Things (IoT) to tackle the road safety challenges arising from the increased rate and volume of traffic due to population growth. In order to enhance road safety and efficiency, the IoV devices enable real-time data transmission and analysis. The proposed multi-tier framework tracks vehicle and roadside unit (RSU) data, encompassing road traffic conditions and vehicle data. The framework integrates vehicles, road traffic, weather conditions, and external factors. On a cloud-based control server, the proposed Spatio-Temporal Conv-Long Short-Term Memory Autoencoder (STCLA) framework deals with and analyzes the resulting data. This research addresses road safety on the Internet of Vehicles via DL. It proposes a novel framework for real-time accident prevention and prediction, demonstrating its effectiveness and potential impact. In a year-long research in Hubei Province, China, data from two road segments demonstrated a substantial boost in predictive accuracy, achieving an Area Under the Receiver Operating Characteristic Curve (AUROC) score of 0.94.

1. Introduction

The volume of vehicles on roads around the world has been rising substantially as the worldwide population has expanded. There continues to be one leading cause for the improvement in traffic size and density: the growth in the volume of vehicles. This increase in crime in vehicular traffic not only extends to the responsibility that is already being imposed on the existing systems but also increases the probability of accidents and the probability of a traffic jam. On top of that, an increase in the volume of vehicles on the roadway has culminated in prolonged journey times, increased usage of fuel, and more emissions of carbon dioxide, all of which have been deemed to be sustainability concerns that are of significant significance.

Meanwhile, there is a direct correlation between an increase in traffic size and an increase in the number of road accidents that cause deaths and injuries on every continent [1]. Accidents with a high rate of frequency result in growing economic and social problems, which include the costs of medical treatment and the loss of worker efficiency.

As a result, it is not only essential but also vital to find an effective tool that might help control the motion of vehicles in order to minimize the sheer volume of accidents [2]. During the past few decades, the Internet of Things (IoT) has developed as an Intelligence Transportation System (ITS) that has extended throughout several types of business

sectors, such as medical services and manufacturing goods, as well as smart homes, home automation sectors, and urban planning. By helping devices that are connected to collect, exchange, and manipulate data in real-time, it has reinvented the methods in which the general population connects with the environment surrounding people [3]. This has succeeded in improving efficiency, productivity, and safety when connecting with the rest of the world. The Internet of Vehicles (IoV) is a focused area of the IoT that has been developed for vehicles as a result of the introduction of IoT technology, which has had a significant effect on the field of vehicular networks. Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Pedestrian (V2P) are all allowed to share information with another vehicle in a reliable method because the IoV, which connects the vehicles with a range of electronic devices, sensors, and networks [4].

The implementation of autonomous and connected vehicles assists in the advancement of networks that have the probability to successfully traffic control statistics, monitor traffic in real-time, minimize accidents, and present autonomous rescue services [5]. They also predict probable risks, improve traffic flow, and enhance the complete road trip [6]. This is achieved by blending sophisticated data analytics and Machine Learning (ML) frameworks with the IoV platform.

Applying ML and DL algorithms to IoV technology has enhanced road safety with automated forecasting and intelligent decision-making

E-mail address: xuwei20241@126.com.

<https://doi.org/10.1016/j.ijin.2024.05.002>

Received 6 March 2024; Received in revised form 1 May 2024; Accepted 4 May 2024

Available online 7 May 2024

2666-6030/© 2024 The Author. Published by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

[7]. By monitoring massive amounts of data from vehicles, transportation systems, and traffic flows, these cutting-edge algorithms predict risks, improve traffic flow, and minimize accidents [8]. Past evidence enables ML and DL algorithms to predict emergencies and locations like accidents and hazardous road conditions. The simulations support the traffic management system or function to avoid accidents with this technique. These frameworks permit dynamic traffic management systems that actively respond to changing conditions, minimizing traffic and boosting disaster actions [9]. IoV-based accident prediction techniques meet multiple issues, necessitating more secure and accurate DL approaches.

The heterogeneity and high volume of traffic data impose analytical challenges that prevent existing models from addressing the intricate changes in road traffic and triggering accidents [10]. Conventional models attempt real-time data analysis and prediction, decreasing how successful they are in dynamic traffic situations. Existing models fail to deal with traffic data's spatiotemporal and geographical correlations, vital for accident prediction [11]. They caused attention to develop a framework that can employ high-dimensional data to predict accidents efficiently and precisely.

IoV design and DL methods are implemented in this article in order to enhance road safety with an Accident Prediction and Prevention System. Starting with the end layer, the proposed multi-layer system collects data from vehicles with connectivity and Roadside Units (RSUs) with real-time vehicle and traffic sensors. The middle layer gateway units aggregate and preliminary processing sensor data and transmit it to a cloud-based server, where the proposed Spatio-Temporal Conv-LSTM Autoencoder (STCLA) model predicts accidents. For detailed prediction, the framework involves vehicles, road traffic, weather conditions, and external factors like festivals and local calamities. Principal Component Analysis (PCA) is implemented to determine the best key features from data collected preceding the model's training. More Effective predictive accuracy is the objective of the STCLA framework that has been presented. This is performed by taking into account weather and vehicle factors and adapting to real-time conditions.

1.1. Research motivation

The ever-growing vehicle volume and traffic due to population progression in current safety and efficiency concerns. To tackle these issues, integrating the IoT into IoV emerges as a promising solution. IoV facilitates real-time data transmission and analysis, enabling proactive measures for prevention of accident and optimization of traffic. By linking vehicles and infrastructure elements, such as traffic lights and road sensors, IoV enhances road safety and efficiency. This integration not only improves transportation systems but also paves the way for smart and sustainable mobility solutions amid urbanization and motorization trends, addressing the evolving needs of modern society.

1.2. Research contribution

- Develop an DL Assisted Accident Prediction and Prevention System for IoV.
- Formulate a multi-tier framework integrating vehicle, weather, road traffic, and external factors data.
- Utilised a STCLA framework on a cloud-based control server.

The paper is organized in the following method: The overview of the literature is laid out in Section II, the proposed framework is described in Section III, the results of the analysis are shown in Section IV, and the work comes to an end in Section V.

2. Literature review

For the intention of performing the traffic analysis, the researchers applied the Federated Learning (FL) paradigm. The FL is a federated ML

approach that provides anonymity of data and serves as the basis for training algorithms on several edge devices [12]. The study has exposed the flaws of centralized ML approaches and indicated that the FL is a feasible option that could improve the analysis of vehicular traffic and the safety of vehicles. They have addressed the limitations of existing V2P communication systems in terms of pedestrian trajectory uncertainty. They introduced a danger index, used Long Short-Term Memory (LSTM) networks for trajectory prediction, and presented a V2P collision risk warning method that had achieved accuracy in prediction of about 98 % [13].

[14] have proposed a novel method by utilizing YOLOv4 for real-time object detection and future vehicle position prediction. They integrated vehicle motions in their ML framework to avoid accidents, and the model they developed predicted how vehicles moved more accurately than others. A GA-optimized Neural Network-based Collision Prediction model [15] was developed. They designed IoV rear-end collision prevention. They optimized the network's settings, reviewed collision risks applying V2I and V2V communications, and increased prediction accuracy using DL and GA.

Real-time object detection and motor vehicle motions were used to predict vehicle locations with YOLOv3 [16]. They boosted IoV dynamic prediction and collision prevention. They proved that a real-time collision avoidance model can be developed without all vehicles' wireless communications. Driving vehicle safety hypotheses that depend exclusively on vehicle speeds have been investigated [17]. The algorithm they used evaluated weather conditions, drivers, vehicles, and data on traffic. A Hidden Markov Model (HMM) has been employed in order to enhance the accident prediction method's sensitivity and precision using these data.

The research team designed a III-layered system integrating MEMS and the IoT to identify vehicular motion during accidents [18]. They identified accidents by MDTW. In experiments analyzing the prediction accuracy, their method outperformed traditional algorithms like SVM and KNN [19]. employed the big data for real-time traffic analysis and accident prediction within VANET networks. They used a method that calculates the traffic density and average speed that can be used to predict accident risks instantaneously [20]. have addressed the constraints of traditional ML in traffic accident prediction by proposing a CNN-based algorithm that autonomously detects data features with high accuracy. Their model had better performance by reducing loss and improved prediction accuracy.

The research gap lies in the essential for a inclusive technique to traffic examination and accident forecast within the Internet of Vehicles (IoV). While previous studies addressed numerous features like Federated Learning (FL) and real-time object detection, they lacked integration of multiple factors. The proposed Spatio-Temporal Conv-LSTM Autoencoder (STCLA) framework aims to bridge this gap by analyzing vehicle, road traffic, weather conditions, and external factors on a cloud-based server, offering a holistic solution for real-time accident prevention and prediction in IoV.

3. Proposed methodology

3.1. Architecture

This work proposes an IoV-based Accident Prediction and Prevention System that is based on a multi-tier model. The Multitier model architecture, as presented in Fig. 1, collects data from vehicles and RSU and processes and analyzes that data to predict road accidents using DL techniques. The layer-wise description of the proposed model is explained below:

A. Bottom Tier - Vehicles and Roadside Units (RSUs): The bottom tier acts as a data collection layer that collects data from the vehicles equipped with On-Board Diagnostics (OBD) readers, GPS devices, and multi-sensor arrays generate data about the vehicle's condition,

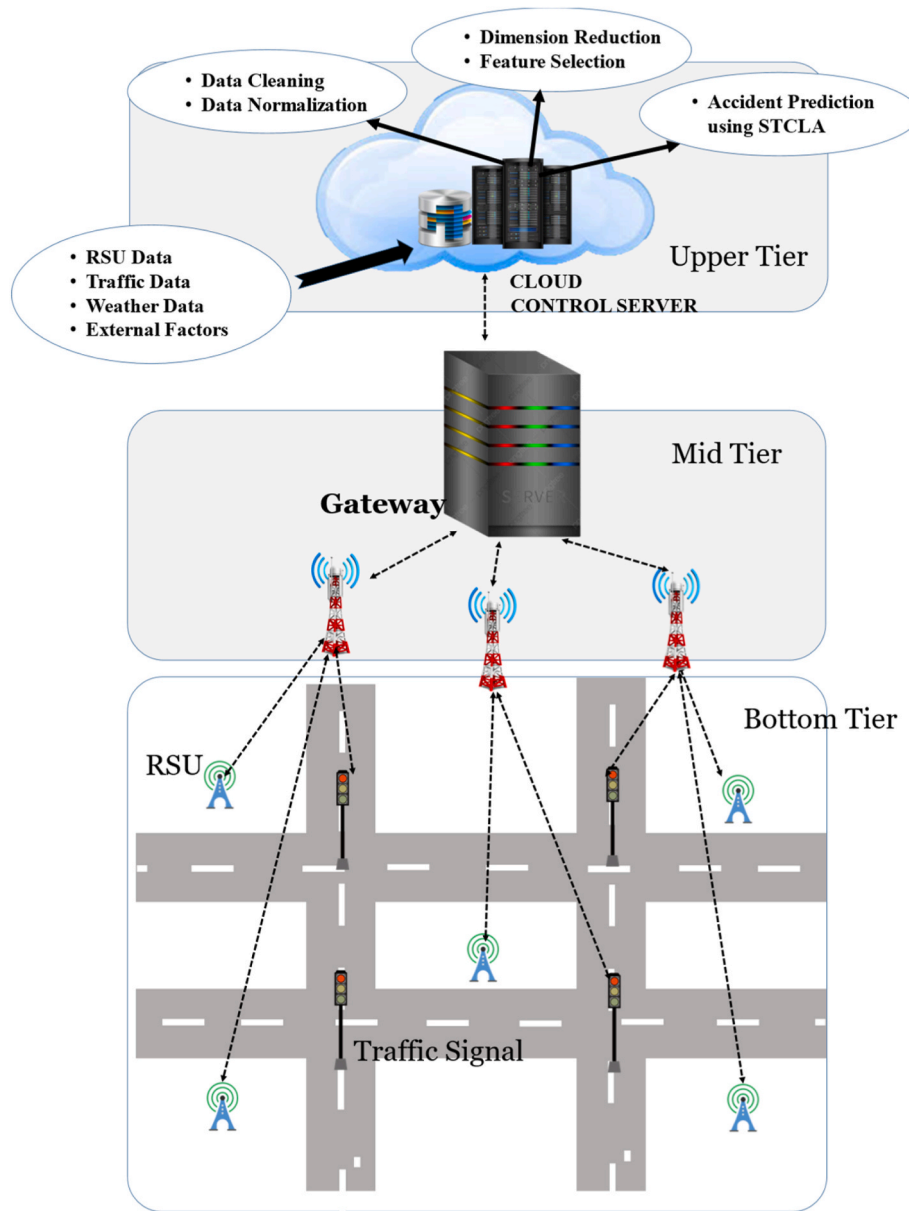


Fig. 1. Proposed architecture.

location, and immediate surroundings. Further, the RSUs also provide data about traffic signals, queue lengths, waiting times, and traffic volumes.

- B. **Mid-Tier - Gateway Units:** Data from the bottom tier are transmitted to gateway units in the mid-tier. This unit processes data from both vehicles and RSUs at a centralized point. Furthermore, this layer of processing performs the preliminary set of preprocessing operations executed on the data collected by the sensors. The procedures encompass data cleaning and normalization. After the data has been aggregated, it becomes available to the next layer.
- C. **Upper Tier-Control Server in the Cloud:** The mid-tier results in data that has been processed to a cloud-based control server web hosting a suggested DL model. The server on which it runs connects vehicular and traffic data with weather conditions, traffic statistics, and external factors like festivals and conventions. This extensive data set is processed by the ML algorithm to find correlations and predict accidents. This multiple-step cloud prediction and prevention system involves data collection, preprocessing, Feature Selection

(FS), hypothesized DL-based prediction model, and prevention system:

3.2. Data collection

Data from multiple sources is used in the proposed accident prediction framework for accurate accident prediction. The following section describes each data source for the intended design. Based on data type, the dataset is divided into subsets of data.

- (i) **Vehicle Data:** This dataset collects vehicle-specific information through:
- **OBD Reader:** A visual representation of the vehicle's present state can be accessed with the help of an On-Board Diagnostics reader, which collects data such as the vehicle's speed, the engine's rotations per minute (RPM), the level of fuel, the battery status, and the engine condition.

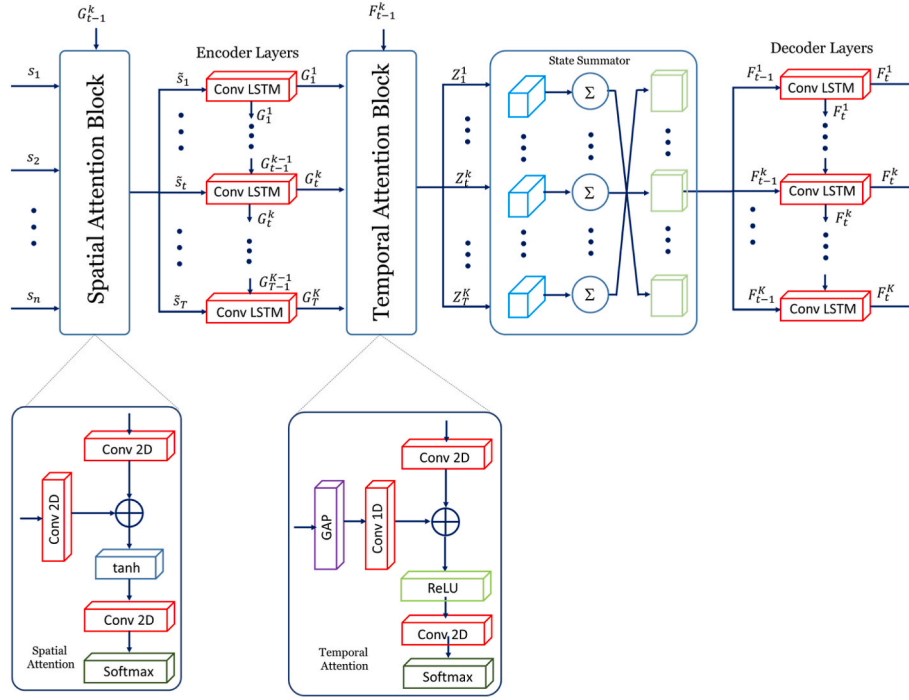


Fig. 2. STCLA prediction model.

- **GPS Reader:** A GPS reader captures location coordinates, time, and altitude, offering spatial and temporal data that is essential for situational analysis.
 - **Multi-sensor Devices:** Devices equipped with radar and lane-change sensors detect the vehicle's immediate external conditions, aiding in assessing the driving environment.
- (ii) **RSU Data:** RSUs provide additional contextual information such as signal timing, vehicle queue lengths, wait times at intersections, and overall traffic volume. This data complements the vehicle's sensor data by adding an infrastructure perspective, allowing for a more comprehensive understanding of the vehicle's environment concerning traffic control measures and congestion levels.
- (iii) **Weather Data:** The weather dataset is tasked with collecting environmental conditions:
- **Temperature:** Records the current temperature, which can affect road conditions.
 - **Humidity:** Monitors moisture levels in the environment, affecting visibility and road surface state.
 - **Precipitation:** Measures rainfall or snowfall, impacting road safety and vehicle handling.
 - **Wind:** Notes wind speed and direction, which can influence driving stability.
 - **Visibility:** Measures clarity of the driving environment, impacting driver response and decision-making.
 - **Pressure:** Monitors barometric pressure for weather prediction.
 - **Cloudiness:** Observes sky conditions, affecting lighting and visibility.
- (iv) **Traffic Data:** This dataset focuses on real-time traffic conditions:
- **Accident Reports:** Gathers data on recent traffic incidents to identify high-risk locations.
 - **Road Closures:** Collects information on road closures that may affect traffic flow.
 - **Road Conditions:** Include details such as speed bumps, dividers, potholes, etc.,
 - **Maintenance and Construction Activity:** Tracks details on ongoing road work, diversions, and changes in traffic patterns.

- (v) **External Factors Data:** This dataset integrates data on socio-economic conditions:

- **Holidays:** Identifies dates that have been predicted to impact the actions of drivers and the overall size of traffic.
- **Weekday Variations:** The aim of this is to collect data on average traffic statistics in order to account for regular variations.
- **Local Events:** This feature collects data about actions that have been predicted to boost traffic or alter standard routes.

3.3. Data processing and feature selection

The procedure of processing the data that has been collected into a format that is appropriate for further investigation using the DL models is addressed in the Data Processing method. Data types from multiple sources have been processed independently. Data cleansing from OBD readers and GPS devices eliminates anomalies that do not match vehicle operation histories or physical limits. This data type normalizes for vehicle-specific features, providing cross-make and type analyses.

Weather-related information is filtered by addressing unread data and verifying for sensor anomalies. The normalization process transforms data to an accepted size, particularly for weather conditions such as wind speed, which is presented in multiple measurements or scales. Traffic data needs cleansing to remove duplicate reports and integrate data from different entities. This data normalizes reports of events into an accepted format for comparison and analysis. Standard coding normalizes the level of accident and road closure data. External factors are cleansed and divided, and traffic patterns are standardized for impact scales. PCA is implemented to reduce the dataset for FS in the proposed framework. PCA minimizes FS and retains the PC, which reflects the most essential data change. The PCA algorithm prevents adapting and maintains the predictive model reliable and applicable to others.

3.4. Proposed Spatio-Temporal Conv-LSTM Autoencoder (STCLA) prediction model

In the proposed learning model, for each point in time within the sequence of spatio-temporal data $s_t = \{s_1, s_2, \dots, s_t\}$, the initial step in-

involves weighting these inputs by the spatial attention coefficients $\gamma_t = (\gamma_t^1, \dots, \gamma_t^i, \dots, \gamma_t^d)$. This operation generates the modified inputs $\tilde{S}_t = (S_t^1 \cdot \gamma_t^1, \dots, S_t^i \cdot \gamma_t^i, \dots, S_t^d \cdot \gamma_t^d)$. The spatial attention mechanism calculates each attention co-efficient γ_t^i constructed on the feature extracted and the prior layer's hidden state $G_{(t-1)}^1$ from the encoder. These weighted inputs, $\tilde{S}_t$, for every moment, are then introduced into the ConvLSTM unit within the encoder's first layer, where $\tilde{S}_t^i$ belongs to $R^{M \times N \times d}$. Following the encoder layers that process the input sequence, the hidden states of the encoder layer $G = (G^1, \dots, G^k, \dots, G^K)$ are forwarded to the temporal attention module. This module calculates the temporal attention weights δ_t^i utilizing the preceding hidden state of the decoder $F_{(t-1)}^k$, encapsulating the input data as a weighted aggregate of the encoder's hidden states over all points in time. The outcome, a context vector $Z = (Z^1, \dots, Z^k, \dots, Z^K)$, serves as the input for the State Summator unit, where each $Z^k = (Z_1^k, \dots, Z_t^k, \dots, Z_T^k)$. The State Summator aggregates the states for the time sequence and inverts the sequence of layers to compile the feature set $F_{(t-1)} = (F_{(t-1)}^1, \dots, F_{(t-1)}^k, \dots, F_{(t-1)}^K)$.

This paper employs the STCLA learning model to control multi-source high-dimensional data. For a series of n spatiotemporal inputs denoted as $s_n = \{s_1, s_2, \dots, s_n\}$ where each input S_t is part of the space $R^{M \times N \times n}$, the spatial attention mechanism [Qin et al. (2017) and Zhang et al. (2018)] is employed. At given 't' timestep, the input spatiotemporal series $S = \{S_{t(i=1)}^i\}_{i=1}^d$ is evaluated using the attention module together with the hidden state $G_{(t-1)}^1$ resulting from the encoder's first layer.

Using these two inputs, the attention module computes the energy matrices for each $i = 1, \dots, d$, EQU (1).

$$E^i = V_G \cdot \tanh(W_G \cdot G_{(t-1)}^1 + U_G \cdot S^i) \quad (1)$$

The attention weights $\gamma_{(k,l)}^i$ are then determined as follows: EQU (2)

$$\gamma_{(k,l)}^i = p(\text{att}_{(k,l)} | S^i, G_{(t-1)}^1) = \frac{\exp(E_{(k,l)}^i)}{\sum_k \sum_l \exp(E_{(k,l)}^i)} \quad (2)$$

Consequently, the resultant modified inputs are attained by applying the attention weights to the original input sequence, EQU (3)

$$\tilde{S}_t = \gamma \odot S_t \quad (3)$$

where $\gamma \in R^{M \times N \times d}$, $E^i \in R^{M \times N \times 1}$, and $S^i \in R^{T \times M \times N}$, it's noted that the convolution operations are performed in 2-D, where $W_G \in R^{K \times K \times m \times Q}$, $U_G \in R^{K \times K \times d \times Q}$, and $V_G \in R^{K \times K \times Q \times 1}$. Here, m, d, Q , and K denote the dimensions of the hidden state, input, attention mechanism, and convolution kernel, respectively. Following the convolution operations, the 2-D energy matrices for each feature are generated; using this, the attention matrix is calculated using a SoftMax operation along the feature dimension. Each element within the matrix A signifies the degree of attention allocated to the features of that particular cell. The output from the spatial attention mechanism, $\tilde{S} = (\tilde{S}_1, \dots, \tilde{S}_t, \dots, \tilde{S}_T)$, is then input into the ConvLSTM units across each encoder layer [21]. The ConvLSTM unit has distinct hidden states G_t^k and F_t^k for each layer where k indicates the layer index. The resulting samples from the input sequence where $t > T_{in}$ for all G_t^k and F_t^k is forwarded to the temporal attention module. The decoder then selects the most relevant hidden states across all time steps by assigning the attention weights specifically to each encoder hidden state at a given time t based on the preceding decoder hidden state $F_{(t-1)} \in R^p$ and the cell state from the ConvLSTM unit $G'_{(t-1)} \in R^p$ using EQU (4) and EQU (5):

$$Z_t^i = V_f^\top \tanh(W_f[F_{(t-1)}; G'_{(t-1)}] + U_f H_i), 1 \leq i \leq T \quad (4)$$

and

$$\delta_t^i = \exp(Z_t^i) / \left(\sum_{j=1}^T \exp(Z_t^j) \right) \quad (5)$$

where the concatenation $[F_{(t-1)}; G'_{(t-1)}] \in R^{2p}$ combines the previous hidden state and cell state from the ConvLSTM unit. The parameters $V_f \in R^m$, $W_f \in R^{m \times 2p}$, and $U_f \in R^{m \times m}$ are to be learned. Bias terms are excluded for simplicity. For the prediction, the weightage of the hidden state of the i -th encoder is measured by the attention weight δ_t^i . The attention mechanism then measures Z_t . The context vector, which is the overall weighted sum of all encoder hidden states $\{G_1, G_2, \dots, G_T\}$, EQU (6).

$$Z_t = \sum_{i=1}^T \delta_t^i G_i \quad (6)$$

It's crucial to recognize that the context vector Z_t varies at every time step. After processing each entry in the input sequence, that is, when $t > T_{in}$, the context vector Z_t , along with each layer's hidden state Z_t^k . It is then forwarded to the state summation, which is the completion of the encoding phase.

The decoder layer is employed to interpret the encoded data and is composed of ConvLSTM units arranged in stacks, mirroring the configuration of the encoder. To mitigate degradation in performance for inputs of extended length, a State Summator mechanism is implemented, which aggregates the hidden state's context vectors Z_t^k for every encoder layer across the temporal dimension, thereby extending the gradient flow back to the initial time step. Following findings from Refs. [22,23], which noted improved outcomes from inversely aligning the encoder's layer hidden states, this symmetry is adopted within our model's design, maintaining a similar layer count between the encoder and decoder. Consequently, as the layer count ascends within the encoder, the dimensions of the hidden states start to decrease and increase for the decoder [24].

In this architecture, the encoder-decoder framework is followed by the process of encoding dense representations and decoding them to expand the representation. The context matcher mechanism can be outlined as follows: EQU (7)

$$F_t^{K-k+1} = \sum_{i=t-T}^t Z_i^k \quad (7)$$

Here, $F_t^{K-k+1} \in R^{M \times N \times m}$ represents the hidden state of the decoder, where K denotes the total number of layers, and $k \in \{1, \dots, K\}$. The decoder then utilizes this output along with $F_{(t+1)}$ for the ensuing prediction. This iterative prediction process persists until $t = T$. Both sets of hidden states are denoted as F_t . The terminal layer of the decoder is tasked with generating the inputs for the output convolutions. These output convolutions enable the utilization of higher dimensions in the encoder's first layer, allowing for the generation of final features with the required dimensions. By employing the selections made in the preceding time step, the decoder facilitates the creation of selections of any sequence length. This design is observed to prevent overfitting and enhance generalization capabilities.

Inputs:

- $s_n = \{s_1, s_2, \dots, s_n\}$: Series of n spatiotemporal inputs
- Parameters: $W_G, U_G, V_G, W_f, U_f, V_f$ - Model parameters

Output: Predicted sequence of potential accident locations/conditions

Begin

- 1 **Initialize** the encoder and decode the ConvLSTM units with random weights.
- 2 **Define** the number of layers K for the encoder and decoder to ensure symmetry.
- 3 **For Each** input sequence S_t in s_n **Do**:
 - **Spatial Attention Mechanism:**
 - **For Each** time step t and feature index i **Do**:
 - **Calculate** energy matrix E^i , EQU (8)

$$E^i = V_G \cdot \tanh(W_G \cdot G_{(t-1)}^1 + U_G \cdot S^i) \quad (8)$$
 - **Determine** attention weights $\gamma_{(k,l)}^i$ for each spatial location (k, l) , EQU (9)

$$\gamma_{(k,l)}^i = \exp(E_{(k,l)}^i) / \left(\sum_{k=1}^M \sum_{l=1}^N \exp(E_{(k,l)}^i) \right) \quad (9)$$
 - **Apply** attention weights to get modified inputs, EQU (10)

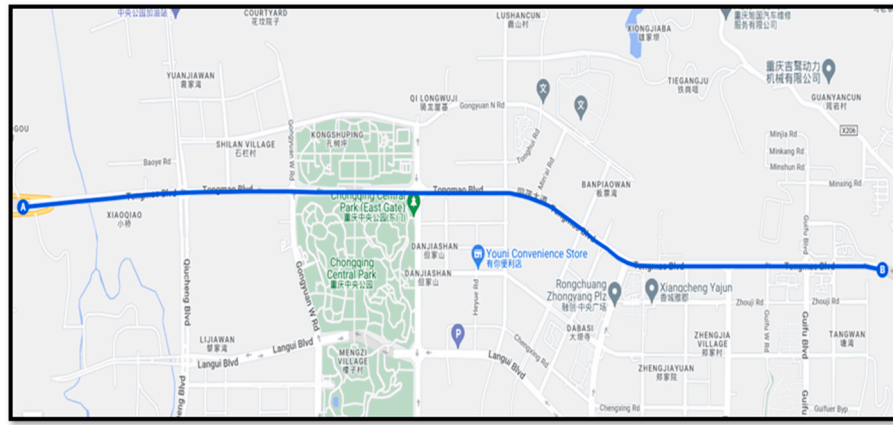
$$\tilde{S}_t = \gamma \odot S_t \quad (10)$$
 - **Encoding with ConvLSTM:**
 - Pass $\tilde{S}_t$ through encoder ConvLSTM units, updating hidden states G_t^k and cell states F_t^k .
 - **Temporal Attention and Decoding:**
 - After processing the entire sequence, employ temporal attention:
 - **For Each** encoder, a hidden state G_i , calculate attention weight δ_t^i , EQU (13)

$$Z_t^i = V_f^T \cdot \tanh(W_f [F_{(t-1)}; G'_{(t-1)}] + U_f \cdot H_i) \quad (13)$$

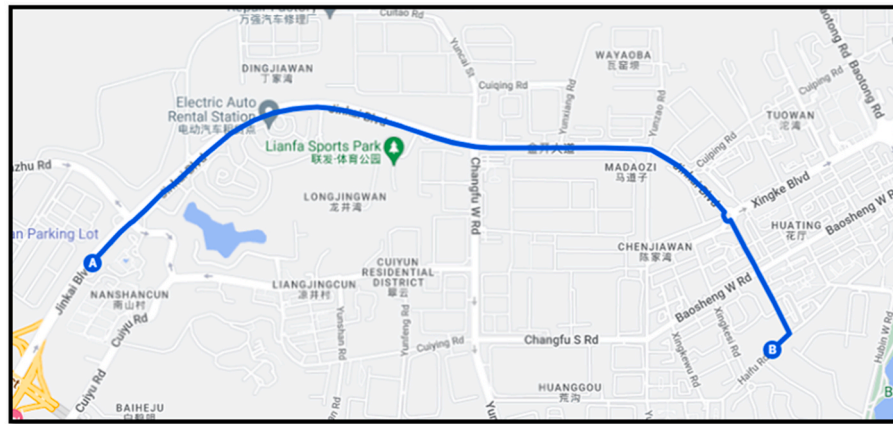
$$\delta_t^i = \exp(Z_t^i) / \left(\sum_{j=1}^T \exp(Z_t^j) \right)$$
 - **Compute** context vector C_t as the weighted sum of encoder hidden states, EQU (14)

$$C_t = \sum_{i=1}^T \delta_t^i \cdot G_i \quad (14)$$
 - Pass context vector C_t through decoder ConvLSTM units to generate predictions.
 - **Output Generation:**
 - The final layer of the decoder processes predictions to produce an output sequence.
 - Use output convolutions to adjust dimensions as needed.
 - 4 **Repeat Steps** for each batch of input sequences in the training dataset.
 - 5 **Training:**
 - Optimize model parameters using backpropagation and an appropriate loss function.

End



(a) Tong Mao Blvd Road



(b) Jinkai Blvd Road

Fig. 3. Research area.

4. Experiment analysis

4.1. Research area and data collection

For the implementation of this prediction model, two road stretches from Hubei Province, China, were selected. The segments are Tong Mao Blvd Road (Fig. 3 (a)) and Jinkai Blvd Road (Fig. 3 (b)), each stretching about 4 km. Both stretches had seven intersections and six intersections, respectively. The data were collected during a period between June 2022 and May 2023 from RSU units placed in the road stretch and in each of the intersection points that collected the vehicle data and traffic data. The accident data and road data were provided by the Ministry of Public Security, specifically through its Traffic Administration Bureau (TAB). The weather data was sourced through the China Meteorological Administration (CMA), and the other external factors related to holidays, festivals, and events are sourced through respective websites.

The TAB provided each accident event, including the accident time, place, and severity. The accident resulting from driver inebriation is omitted as it is inappropriate for real-time traffic and signal attributes. The data after such omission resulted in a total of 83 accidents (44 on Tong Mao Blvd Road and 39 on Jinkai Blvd Road). The RSU data provided both the vehicle data (Speed) and the traffic volume. The vehicle data are transmitted to RSU that give the details about each vehicle speed, and the RSUs placed at the stretches collect the traffic volume of the entire stretch at a given point of time, particularly in our study, a 15-min aggregate traffic volume of the stretch is managed through the installed RSUs. The RSUs installed in the traffic signals collect data

related to vehicle queue length and waiting time at the signal. The entire stretch has a two-way traffic flow, so both features are included. The research area comprises Tong Mao Blvd Road and Jinkai Blvd Road, as illustrated in Fig. 3. Tong Mao Blvd Road and Jinkai Blvd Road are critical segments under study. Analyzing these road sections provides insight into traffic patterns and facilitates the development of effective accident prediction and prevention strategies within the designated area.

The weather features collected from CMA included visibility, humidity, precipitation, wind speed, and temperature. The prediction and prevention model focuses on a smaller 5-min window for effective prevention probability. So, the features sourced and generated all fall within the time window of 5-min intervals. All the statistical features are shown in Table 1. Overall, the first stretch road had a sample size of 326865 samples, and the second stretch had around 384051 samples, resulting in a total sample size of 710916 collected for one year.

A. Data Processing: Using the specific statistical measures for each feature, value-based techniques were employed to ensure data accuracy, reliability, and compatibility across diverse sources. The preprocessing step for each data type is illustrated below:

4.1.1. RSU data processing techniques

- **Vehicle Speed:** Data outliers beyond the established range (20 km/h to 100 km/h) are identified and reviewed for accuracy.

Table 1
Dataset description.

Type	Feature	Description	Mean (Std)	(Min, Max)
RSU Data	Vehicle Speed	Average speed of vehicles recorded by RSUs, measured in km/h.	45 km/h (15 km/h)	(20 km/h, 100 km/h)
	Traffic Volume	The count of vehicles passing a point for a specified time is represented as a count.	200 vehicles (50)	(100, 400)
	Vehicle Queue Length	Length of vehicle queue at signals detected by RSUs, measured as a count of vehicles.	10 vehicles (5)	(0, 20)
	Waiting Time	Waiting time at signals recorded by RSUs, measured in seconds (sec).	120 s (30 s)	(60 s, 180 s)
Weather Data	Visibility	Visibility conditions were reported by CMA and measured in kilometers (km).	8 km (2 km)	(0 km, 10 km)
	Humidity	Humidity levels reported by CMA are measured as a percentage (%).	60 % (10 %)	(40 %, 80 %)
	Precipitation	Precipitation rates reported by CMA, including rain or snow, are measured in millimeters (mm).	5 mm (10 mm)	(0 mm, 50 mm)
	Wind Speed	As reported by CMA, wind speed is measured in kilometers per hour (km/h).	15 km/h (5 km/h)	(0 km/h, 25 km/h)
Accident Data	Temperature	The ambient temperature, as reported by CMA, is measured in degrees Celsius (°C).	20 °C (5 °C)	(-5 °C, 35 °C)
	Time	Timestamp when the accident occurred, represented as time.		
	Location	The specific location of the accident occurrence is represented as a location.		
External Factor Data	Severity	The severity of the accident is categorized into no/minor injury, severe injury, or fatality.		
	Holiday	Indicates whether the day is a holiday, represented as a binary (Yes/No).		
	Festival	The occurrence of festivals within the study period is represented as a binary (Yes/No).		
	Events	Significant events affecting traffic patterns are represented as a binary (Yes/No) (see Fig. 2).		

- **Traffic Volume:** Traffic counts that are falling outside the typical range (100–400 vehicles) are scrutinized for anomalies.
- **Vehicle Queue Length:** Queue lengths at signals are maintained within the observed range (0–20 vehicles).
- **Waiting Time at Signals:** Waiting times are standardized across intersections, and the values outside the 60-s to 180-s range are corrected to ensure consistency.

4.1.2. Weather data processing techniques

- **Visibility:** Visibility readings below 0 km or above 10 km are considered anomalies and subjected to verification.
- **Humidity:** Humidity data is normalized within the 40 %–80 % range.
- **Precipitation:** Precipitation measurements are standardized to reflect actual weather conditions, with readings outside the 0–50 mm range examined for data integrity.
- **Wind Speed:** Wind speeds exceeding the 0–25 km/h range are assessed for accuracy.

- **Temperature:** Temperature data is saved within the range of –5 to 35 °C.

4.1.3. Accident data processing techniques

- **Accident Severity:** Accident reports are categorized into three severity levels: no/minor injury, severe injury, and fatality.

4.1.4. External factor Data Processing techniques

- **Holidays, Festivals, and Events:** The occurrence of holidays, festivals, and events is flagged as binary (Yes/No).

4.2. Feature selection using Principal Component Analysis (PCA)

PCA technique was used for dimension reduction of the collected dataset while retaining as much variability as possible. The first step in PCA involves standardizing the range of continuous initial variables so that each feature contributes equally to the analysis. Given the diverse range of the collected data, each feature is standardized to have a mean of 0 and a standard deviation of 1. Following standardization, the covariance matrix is calculated to understand how the variables in the dataset vary from the mean concerning each other. Fig. 4 shows the matrix that expresses the correlation between each pair of features in the dataset.

The next step involves computing the eigenvalues and eigenvectors of the covariance matrix. Eigenvectors point in the direction of maximum variance in the data, while eigenvalues signify the magnitude of that variance. The eigenvectors with the highest eigenvalues are the principal components of the dataset. Selecting the top eigenvectors based on their corresponding eigenvalues allows the form of a feature vector. This vector is essentially a matrix that acts as a tool to transform the original dataset into a new one where the features are uncorrelated. The number of eigenvectors selected corresponds to the number of dimensions onto which the data will be reduced. Finally, the original dataset is transformed into a new dataset by multiplying the original standardized data matrix by the feature vector. The resulting dataset is a reduced representation of the original, with features (principal components) that are linear combinations of the actual variables.

After applying PCA to the dataset encompassing vehicle data, weather conditions, traffic patterns, and external factors, the analysis yields a set of principal components that capture the most significant variance within the data. Based on the eigenvalues and the cumulative explained variance ratio, we select the top elements that collectively define a substantial portion of the variability (typically above 80 %).

The final list of selected features, represented as principal components, includes:

- **PC1:** A weighted combination of vehicle speed, traffic volume, and queue length, highlighting the interaction between traffic flow and congestion.
- **PC2:** Captures variations in weather conditions, combining visibility, humidity, precipitation, wind speed, and temperature, reflecting the impact of weather on driving conditions.
- **PC3:** Represents the influence of external factors, aggregating the effect of holidays, festivals, and significant events on traffic patterns.
- **PC4:** Reflects a combination of waiting times at signals and specific accident severity categories, indicating potential risk areas.

These principal components serve as the input features for the predictive model, encapsulating the multidimensional aspects of the data in a reduced form that is more manageable and insightful for analysis. By focusing on these composite features, the model leverages the most critical information from the dataset, enhancing its ability to predict and prevent road accidents effectively.

When performing Principal Component Analysis (PCA) for feature

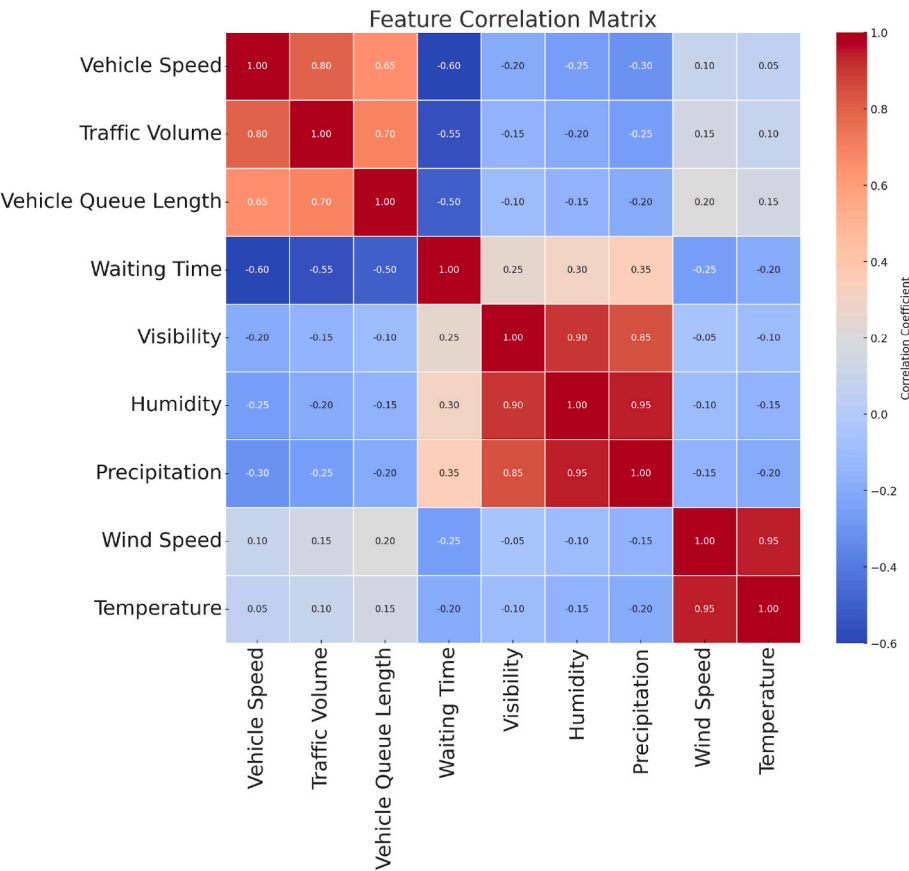


Fig. 4. Feature correlation.

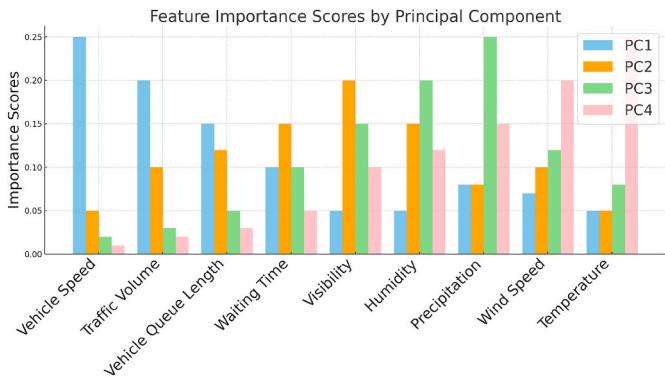


Fig. 5. Feature importance score.

Table 2	
Hyperparameter for the model training.	
Hyperparameter	Value/Range
Number of Layers (K)	4-6 layers
ConvLSTM Units	64, 128, 256 (per layer)
Kernel Size	(3 × 3), (5 × 5)
Activation Function	ReLU (ConvLSTM), Sigmoid (Output)
Dropout Rate	0.2–0.5
Learning Rate	0.001–0.01 (Adaptive)
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Batch Size	32, 64, 128
Epochs	100 (with Early Stopping)
Early Stopping Patience	10–20

selection, each of the Principal Components (PCs) is a linear combination of the original variables, and their coefficients can indicate the importance of each feature in that component. While PCA itself does not directly provide a feature importance score as some ML models do, the square of the coefficients (loadings) in each PC can be interpreted as the contribution of each feature to the variance explained by that component, as shown in Fig. 5.

These scores are based on the squared loadings of each feature within each PC, representing the percentage of the component’s variance that each feature contributes. It’s important to note that the sum of squared loadings for each element equals the eigenvalue of that component, which indicates the total variance explained by the component. The importance scores help in understanding the relative contribution of each feature to the principal components, guiding the selection of features for further modeling or analysis.

4.3. Experiment setup and model training

All the experiments were tested using a system configured with an Intel Xeon Processor, 64 GB of RAM, and an NVIDIA Tesla V100 GPU that is operated on a Linux-based OS. For software, Python 3.8 is used as the programming language with libraries such as Pandas for data manipulation, NumPy for numerical operations, and TensorFlow, along with Keras for the development and training of the ConvLSTM DL models. After feature selection, the dataset is split into 70:30 for training and testing. The proposed model is trained using the parameters listed in Table 2:

The model is validated using metrics such as Area Under the Receiver Operating Characteristic Curve (AUROC), precision, recall, and F1 score.

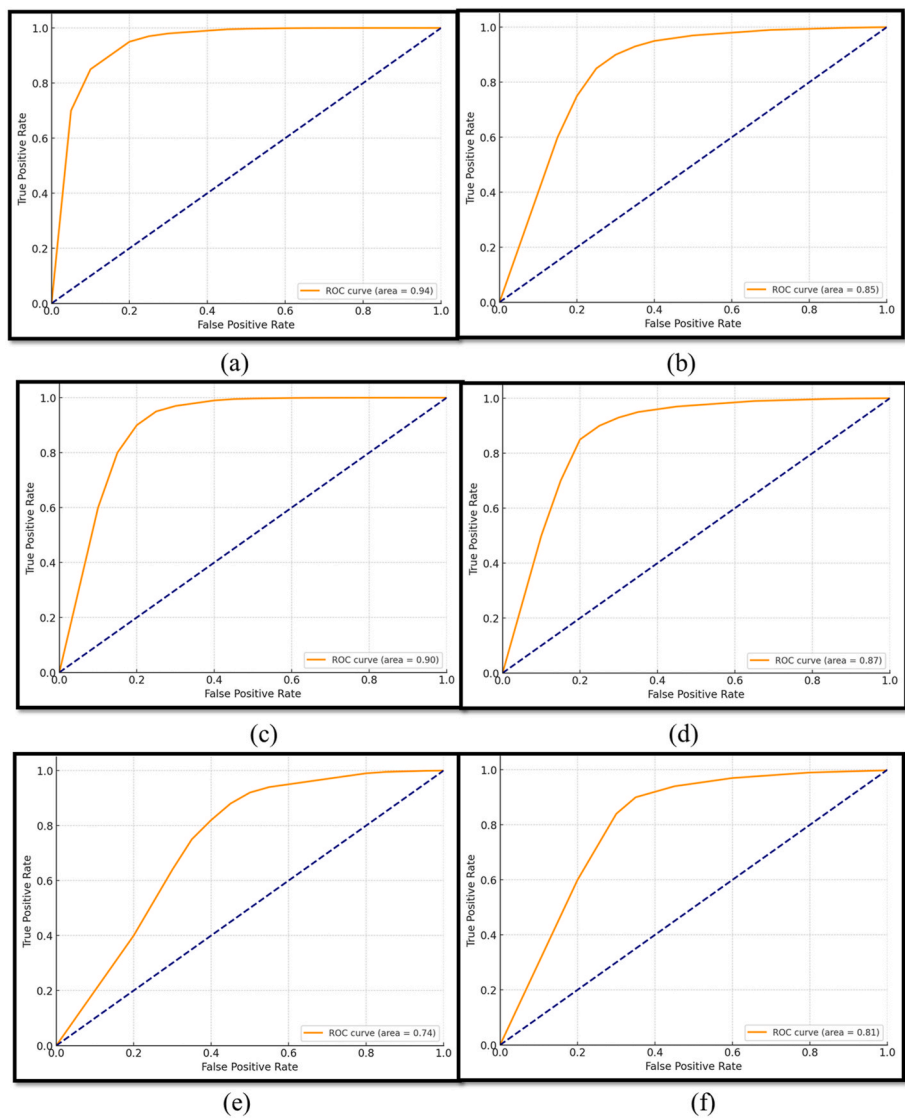


Fig. 6. AUROC results of a) Proposed, b) GBM, c) LSTM, d) AE, e) SVM and f) CNN.

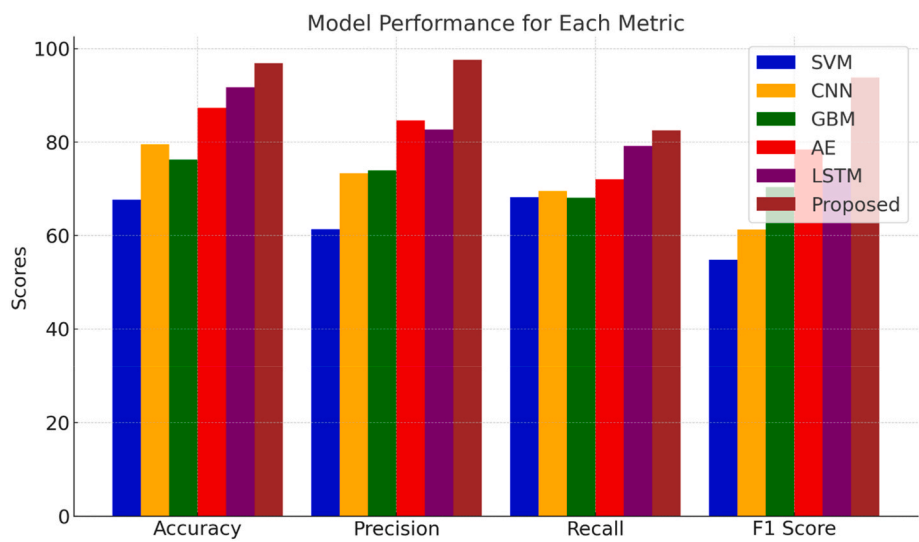


Fig. 7. Performance of compared model vs metrics.

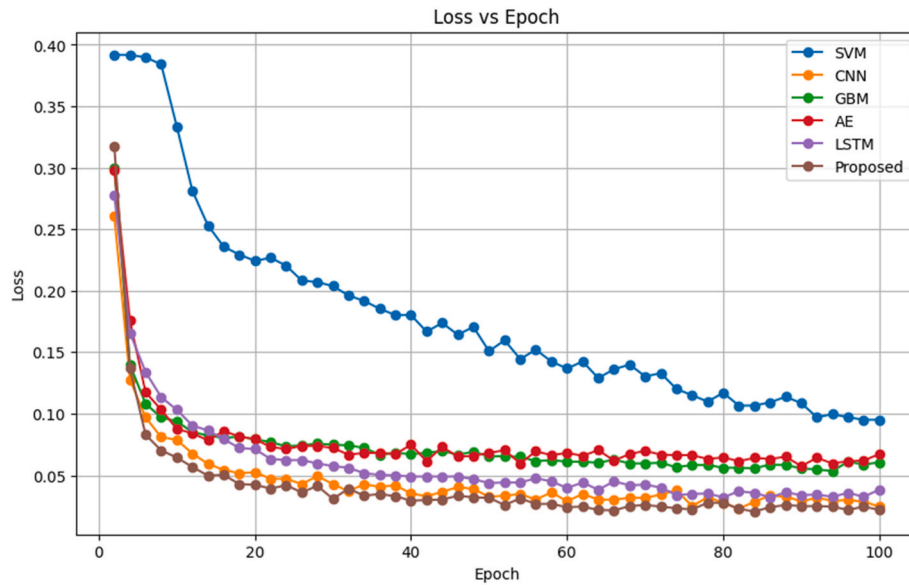


Fig. 8. Loss vs Epoch.

- 1 **Accuracy:** Accuracy measures the proportion of true results among the total number of cases examined. It is calculated as the sum of TP and TN divided by the total number of cases, EQU (15)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

- 2 **Precision (P):** Precision measures the accuracy of the optimistic predictions made by the model, EQU (16)

$$P = \frac{TP}{TP + FP} \quad (16)$$

where TP represents true positives, and FP means false positives.

- 3 **Recall (R) or True Positive Rate:** Recall measures the ability of the model to capture all relevant cases, EQU (17)

$$R = \frac{TP}{TP + FN} \quad (17)$$

where FN represents False Negatives.

- 4 **F1 Score:** The F1 score is the harmonic mean of precision and recall, providing a balance between EQU (18).

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (18)$$

Fig. 6 (a)-(f) displays the AUROC scores of all the compared models. The Spatio-Temporal Conv-LSTM Autoencoder Prediction Model demonstrates high accuracy in accident prediction, as reflected by its AUROC score of 0.943. This score indicates an excellent predictive performance, particularly when compared to other models in the same application. Comparative analysis shows that the Gradient Boosting Machine (GBM) model has an AUROC score of 0.837, and the Long Short-Term Memory (LSTM) network has a score of 0.851. Both are effective for their respective purposes but do not reach the level of the proposed model. An Autoencoder (AE) used in the study shows an AUROC of 0.74, suggesting limitations in accident prediction without additional supervised learning. The Support Vector Machine (SVM) achieved an AUROC of 0.831; though the score is significant, it is still lower than the proposed model. The CNN model, which is better at

processing spatial data, recorded an AUROC of 0.90. Despite its strengths, it does not incorporate the temporal dynamics as effectively as the proposed model. The proposed model has better integration of both spatial and temporal data, which was proven by its better AUROC score than the other models.

Fig. 7 displays the performance of the compared models against different metrics such as accuracy, precision, recall, and F1-score. The proposed model outperforms the other models across all metrics with an accuracy near 97 % and a precision exceeding 97 %; it demonstrates high reliability in its predictions, minimizing both FP and FN. The recall is high but slightly lower than precision, showing it successfully captures most TP instances. The F1-score, which balances precision and recall, is scored high at 93.77 %, confirming the model's robustness (see Fig. 8).

In contrast, the other models show varying performance levels, with SVM having an accuracy of 67.69 %, the lowest among the models. Its precision is 61.37 %, indicating a higher rate of false positives. A recall is 68.14 %, slightly above precision, but the F1-score is 54.77 %, showing SVM's limited effectiveness. CNN improves accuracy to 79.46 %. Precision at 73.34 % and recall at 69.54 % are closely matched, with an F1-score of 61.26 %, suggesting CNN's capability with spatial data. GBM shows accuracy at 76.23 % and precision at 73.89 %, with recall nearly matching CNN. The F1-score of 70.36 % surpasses SVM and CNN, showing better precision and recall balance. AE has an accuracy of 87.32 % and a precision of 84.68 %. The recall is at 72.02 %, with an F1-score of 78.42 %, indicating AE's classification strength, albeit with potential misses on true positives. LSTM shows an accuracy of 91.74 %. Its precision is 82.66 %, with recall at 79.17 %, indicating its capability to identify relevant cases. The F1-score is 74.16 %, slightly lower than AE, suggesting a variance in precision and recall balance.

Fig. 6 evaluates the Binary Cross-Entropy (BCE) loss over 100 epochs for different models aimed at predicting accidents, and a clear differentiation in performance emerges. The SVM model begins with a BCE loss of 0.3917 and concludes at 0.009514, demonstrating a steady decrease. The CNN model, starting from a loss of 0.260394, reduces this significantly to 0.0246, indicating its effectiveness in learning from spatial data rapidly. The GBM model's loss decreases from 0.2994 to 0.0606, showing consistent progress in learning. The Autoencoder (AE) model started at 0.2981 and ended at 0.0672, highlighting its data reconstruction and learning capability. The LSTM model that begins with an initial loss of 0.2777 later reduces it to 0.0384, which reflects its strength in analyzing time-series data. The proposed framework was learned from spatiotemporal information to accurately predict

accidents, beginning with 0.3169 loss and dropping to 0.0219. The results of this evaluation demonstrate that the proposed approach decreases loss most significantly, SVM achieves minor advancement, and LSTM and AE function effectively. CNN's dataset adaptation was fast, while GBM did somewhat well.

5. Conclusion and future work

This work demonstrates an accident predictor and mitigation system that uses deep learning and the Internet of Vehicles to boost road safety. Researchers in Hubei Province, China, found that the model—a multi-tier Spatio-Temporal Convolution-Long Short-Term Memory Autoencoder—was very good at predicting accidents that would happen in the future. It had an Area Under the Receiver Operating Characteristic Curve of 0.94. The research topic focuses on the potential uses of the Internet of Vehicles and machine learning models for real-time traffic tracking and accident prevention, highlighting their pros and cons while paying attention to their flaws, such as an excessive focus on spatial information and an absence of additional data sources. This paper uses federated learning models for scalable and secure techniques, as well as pedestrian and bicycle data for easier navigation.

To adapt to environmental and geographical factors using artificial intelligence tools, further evaluation and research are required.

Data availability

Not Applicable.

Funding acknowledgements

Not Applicable.

CRediT authorship contribution statement

Xu Wei: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] A.A. Musa, S.I. Malami, F. Alanazi, W. Ounaies, M. Alshammari, S.I. Haruna, Sustainable traffic management for smart cities using internet-of-things-oriented intelligent transportation systems (ITS): challenges and recommendations, *Sustainability* 15 (13) (2023) 9859.
- [2] E.O. Oladeji, C. Ezeme, L.A. Baiyewu, M.O. Okunola, S.O. Ogunlade, The catastrophic cost of motorcycle road traffic injuries: experience from a major reference centre in a lower-middle income country, *Injury* 111314 (2024).
- [3] P. Kaledio, J. Oloyede, F. Olaoye, D. Lucas, *The Internet of Things: Transforming Industries, Empowering Connectivity, and Shaping the Future*, 2023.
- [4] T. Aldhanhani, A. Abraham, W. Hamidouche, M. Shaaban, *Future Trends in Smart Green IoV: Vehicle-To-Everything in the Era of Electric Vehicles*, *IEEE Open Journal of Vehicular Technology*, 2024.
- [5] K.N.H. Amlan, M.S. Uddin, T. Mahmud, N.B. Riyan, *IoT, cloud computing, and sensing technology for smart cities*, in: *Intelligent Techniques for Cyber-Physical Systems*, CRC Press, 2023, pp. 267–291.
- [6] B. Deng, J. Nan, W. Cao, W. Wang, Survey on integration of network communication into vehicle real-time motion control, *IEEE Commun. Surv. Tutor.* (2023).
- [7] X. Wang, H. Zhu, Z. Ning, L. Guo, Y. Zhang, Blockchain intelligence for internet of vehicles: challenges and solutions, *IEEE Commun. Surv. Tutor.* (2023).
- [8] S. Dikshit, A. Atiq, M. Shahid, V. Dwivedi, A. Thusu, The use of artificial intelligence to optimize the routing of vehicles and reduce traffic congestion in urban areas, *EAI Endorsed Trans. Energy Web* 10 (2023).
- [9] A. Sohail, M.A. Cheema, M.E. Ali, A.N. Toosi, H.A. Rakha, Data-driven approaches for road safety: a comprehensive systematic literature review, *Saf. Sci.* 158 (2023) 105949.
- [10] H. Taslimasa, S. Dadkhah, E.C.P. Neto, P. Xiong, S. Ray, A.A. Ghorbani, Security issues in internet of vehicles (IoV): a comprehensive survey, *Internet of Things* (2023) 100809.
- [11] V.I. Kontopoulou, A.D. Panagopoulos, I. Kakkos, G.K. Matsopoulos, A review of ARIMA vs. machine learning approaches for time series forecasting in data-driven networks, *Future Internet* 15 (8) (2023) 255.
- [12] K.R. Reddy, A. Muralidhar, Machine learning-based road safety prediction strategies for internet of vehicles (IoV) enabled vehicles: a systematic literature review, *IEEE Access* (2023).
- [13] R. Pan, L. Jie, X. Zhang, S. Pang, H. Wang, Z. Wei, A V2P collision risk warning method based on LSTM in IOV, *Secur. Commun. Network.* 2022 (2022).
- [14] C.C. Chang, Y.M. Ooi, B.H. Sieh, IoV-based collision avoidance architecture using machine learning prediction, *IEEE Access* 9 (2021) 115497–115505.
- [15] C. Chen, H. Xiang, T. Qiu, C. Wang, Y. Zhou, V. Chang, A rear-end collision prediction scheme based on deep learning in the Internet of Vehicles, *J. Parallel Distr. Comput.* 117 (2018) 192–204.
- [16] C.C. Chang, W.M. Lin, C.A. Lai, Using real-time dynamic prediction to implement IoV-based collision avoidance, *Appl. Sci.* 9 (24) (2019) 5370.
- [17] N. Aung, W. Zhang, S. Dhelim, Y. Ai, Accident prediction system based on hidden Markov model for vehicular ad-hoc network in urban environments, *Information* 9 (12) (2018) 311.
- [18] N. Kumar, S.K. Sood, M. Saini, IoV-fog-assisted framework for accident detection and classification, *ACM Trans. Embed. Comput. Syst.* (2023).
- [19] M. Tantaoui, M.D. Laanaoui, M. Kabil, Big data accident prediction system in green networks and intelligent transportation systems, in: *Emerging Trends in ICT for Sustainable Development: the Proceedings of NICE2020 International Conference*, Springer International Publishing, 2021, pp. 121–127.
- [20] H. Zhao, H. Cheng, T. Mao, C. He, Research on traffic accident prediction model based on convolutional neural networks in VANET, in: *2019 2nd International Conference on Artificial Intelligence and Big Data (ICAIBD)*, IEEE, 2019, May, pp. 79–84.
- [21] Y. Qin, D. Song, H. Chen, W. Cheng, G. Jiang, G. Cottrell, A dual-stage attention-based recurrent neural network for time series prediction, *arXiv preprint arXiv: 1704.02971* (2017).
- [22] X. Shi, Z. Gao, L. Lausen, H. Wang, D.-Y. Yeung, W.-k. Wong, W.-c. Woo, Deep learning for precipitation nowcasting: a benchmark and a new model, in: *Advances in Neural Information Processing Systems*, 2017, pp. 5617–5627.
- [23] Q.-K. Tran, S.-k. Song, Computer vision in precipitation nowcasting: applying image quality assessment metrics for training deep neural networks, *Atmosphere* 10 (5) (2019) 244.
- [24] D. Zhang, G. Lindholm, H. Ratnaw Ee Ra, Use long short-term memory to enhance Internet of Things for combined sewer overflow monitoring, *J. Hydrol.* 556 (2018) 409–418.